

ENTHUSIAST ADVANCE COURSE**LEADER PHASE - MED-I (B)****TARGET - NEET MEDICAL 2024**Test Type : **MINOR**Test Pattern : **NEET (UG)****TEST DATE : 21-05-2023****ANSWER KEY**

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	1	3	3	2	2	3	3	4	2	2	2	1	1	3	4	3	4	1	4	1	4	4	1	1	2	3	2	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	1	2	2	4	3	1	4	2	2	1	1	3	2	4	3	3	2	1	2	2	1	2	2	2	4	2	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	2	4	4	1	2	3	4	3	4	4	4	1	3	3	1	2	4	3	2	4	2	2	3	3	4	2	4	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	2	2	2	4	2	1	2	3	3	3	1	2	3	3	2	2	1	3	3	2	2	4	4	3	3	3	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	4	4	2	4	1	3	4	3	1	2	3	3	2	3	1	4	1	2	1	4	3	2	4	2	1	3	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	4	1	3	3	2	2	3	3	3	1	2	3	2	4	4	1	1	4	2	3	3	3	3	4	2	2	2	1	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	2	4	4	2	3	3	4	2	3	2	4	2	2	4	4	3	3	1	3										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**2. **Ans (1)**

$$F = G(3)(1) \left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \dots \infty \right) = 4G$$

3. **Ans (1)**

$$g = \frac{GMr}{R^3} = \frac{G4\pi R^3 \rho r}{3R^3} = \frac{4\pi G \rho r}{3}$$

4. **Ans (3)**

$$F = \frac{dp}{dt} = 8t - 2$$

$$\text{at } t = 1 \text{ sec, } F = 8 - 2 = 6 \text{ N}$$

5. **Ans (3)**

$$I_I = 1 \times 10^{-3} \times 0.25 = 25 \times 10^{-5} \text{ NS}$$

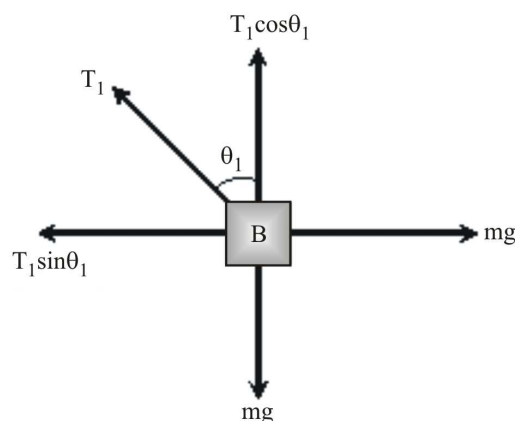
$$I_{II} = \frac{1}{2} \times 2 \times 10^{-3} \times 0.3 = 3 \times 10^{-4} \text{ NS}$$

$$I_{III} = \frac{1}{2} \times 1 \times 10^{-3} \times 1 = 5 \times 10^{-4} \text{ NS}$$

$$I_{IV} = \frac{1}{2} \times 1 \times 10^{-3} \times 1 = 5 \times 10^{-4} \text{ NS}$$

6. **Ans (2)**

From the free body diagram of block B



$$T_1 \cos \theta_1 = mg \dots (i)$$

$$T_1 \sin \theta_1 = mg \dots (ii)$$

by squaring and adding

$$T_1^2 (\sin^2 \theta_1 + \cos^2 \theta_1) = 2(mg)^2$$

$$\therefore T_1 = \sqrt{2}mg$$

7. **Ans (2)**

For a bounded system total energy should be negative.

8. **Ans (3)**

$$V = V_1 + V_2$$

$$V = -\frac{GM}{3\frac{R}{2}} - \frac{G \times 2M}{2R}$$

$$V = -\frac{2GM}{3R} - \frac{GM}{R}$$

$$V = -\frac{5GM}{3R}$$

9. **Ans (3)**

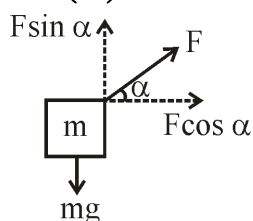
$$a = \frac{T_3}{m_1 + m_2 + m_3}$$

$$a = \frac{60}{10 + 20 + 30} = 1 \text{ m/s}^2$$

$$T_2 = (m_1 + m_2) a$$

$$T_2 = (10 + 20) \times 1 = 30 \text{ N}$$

10. **Ans (4)**



When body leaves the surface

$$F \sin \alpha = mg$$

$$Kt \sin \alpha = mg$$

$$t = \frac{mg}{K \sin \alpha}$$

13. **Ans (2)**

$$R \propto \ell^2 [\text{vol} = \text{constant}]$$

$$\ell_i = \ell \Rightarrow R_i = R$$

$$\ell_f = \ell + \frac{40\ell}{100} \Rightarrow \frac{7\ell}{5} \Rightarrow R_f \Rightarrow \frac{49}{25} R$$

$$\% \text{ change} \Rightarrow \frac{R_f - R_i}{R_i} \times 100$$

$$\% \text{ change} \Rightarrow \frac{\frac{49}{25}R - R}{R} \times 100 = 96\%$$

15. **Ans (1)**

If I is current through ABC

$$I = \frac{5}{21}, V_{AB} = I \times (7 + 7)$$

$$V_{AB} = \frac{5}{21} \times 14 = \frac{10}{3} \text{ Volt}$$

18. **Ans (3)**

$$R_t = R_0(1 + \alpha t)$$

$$5 = R_0(1 + 50\alpha) \dots (1)$$

$$6 = R_0(1 + 100\alpha) \dots (2)$$

$$\text{Solving (1) \& (2)} \Rightarrow R_0 = 4\Omega$$

29. **Ans (2)**

$$\delta_m = A \text{ (given)}$$

$$\sqrt{3} = \frac{\sin\left(\frac{A+A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\sqrt{3} = \frac{\sin A}{\sin A/2} = \frac{2 \sin(A/2) \cos(A/2)}{\sin(A/2)}$$

$$\cos(A/2) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{A}{2} = 30^\circ \Rightarrow \boxed{A = 60^\circ}$$

30. **Ans (1)**

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\Rightarrow \frac{1.5}{v} - \frac{1}{-15} = \frac{1.5 - 1}{30}$$

35. **Ans (2)**

$$\text{VIBGYOR} \rightarrow \lambda \uparrow$$

$$\rightarrow \mu \downarrow$$

$$\rightarrow \theta_c \uparrow$$

$$\sin \theta_c = \frac{1}{\mu}$$

$$\mu \propto \frac{1}{\lambda}$$

SECTION-B

36. **Ans (4)**

If same mass m placed at corners of a pentagon than net force on body of mass 'm' placed at centre is zero

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 0$$

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = -\vec{F}_5$$

$$|\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4| = |\vec{F}_5| = \frac{Gmm}{a^2}$$

now one mass is replaced by m_0 then net force

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5'$$

$$= \vec{F}_5' - \vec{F}_5$$

$$F_{\text{net}} = F_5' - F_5 = \frac{Gmm_0}{a^2} - \frac{Gmm}{a^2}$$

$$F_{\text{net}} = \frac{Gm(m_0 - m)}{a^2}$$

37. **Ans (3)**

$$F_{\text{net}} = f - f = 0$$

So, Tension remains same at all points of the rope and equal to F.

38. **Ans (1)**

From COME :-

$$-\frac{GMm}{R+R} = -\frac{GMm}{R} + \frac{1}{2}mV^2$$

$$\frac{1}{2}mV^2 = \frac{GMm}{2R}$$

$$V^2 = \frac{gR^2}{R} = gR$$

$$\therefore V = \sqrt{gR}$$

39. **Ans (4)**

$$-\frac{GM \times 3}{R} = -54$$

$$\frac{GM}{R} = 18$$

$$V_e = \sqrt{\frac{2GM}{R}} = \sqrt{2 \times 18} = 6 \text{ m/s}$$

40. **Ans (2)**

$$\therefore I = \Delta P$$

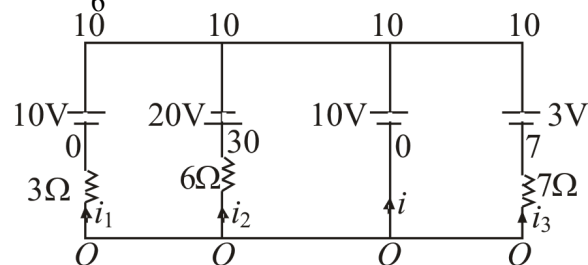
$$\therefore \frac{\pi F_0 \times \frac{T}{2}}{2} = mU$$

$$\therefore U = \frac{\pi F_0 T}{4m}$$

42. **Ans (1)**

$$i_1 = 0$$

$$i_2 = \frac{-30}{6} = -5 \text{ amp}$$



$$i_3 = \frac{0-7}{7} = -1 \text{ amp}$$

$$i_1 + i_2 + i_3 + i = 0$$

$$0 + -5 - 1 + i = 0$$

$$i = 6 \text{ amp}$$

43. **Ans (1)**

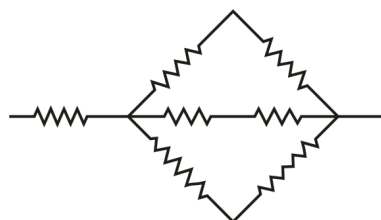
$$R = R_1 + R_2$$

$$\frac{\rho_{\text{eq}}(x_1 + x_2)}{A} = \frac{\rho_1 x_1}{A} + \frac{\rho_2 x_2}{A}$$

$$\rho_{\text{eq}} = \left(\frac{\rho_1 x_1 + \rho_2 x_2}{x_1 + x_2} \right)$$

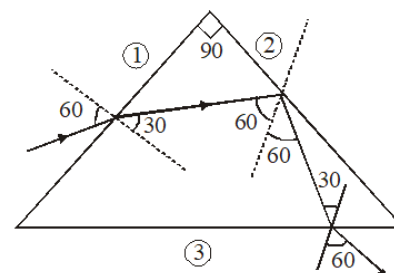
44. **Ans (3)**

Upper half is balanced W.S.B.



$$R_{\text{eq}} = R + \frac{2R}{3} = \left(\frac{5R}{3} \right)$$

49. **Ans (2)**



For Ist surface

$$1 \times \sin 60 = \sqrt{3} \times \sin r_1$$

$$r_1 = 30$$

$$r_2 = 90 - 30 = 60^\circ$$

$$\sin 60 > \sin c \text{ (T.I.R.)}$$

From IIIRD surface

$$\sqrt{3} \times \sin 30 = 1 \times \sin e \quad e = 60$$

$$\delta_1 = 60 - 30 = 30^\circ \text{ C.W.}$$

$$\delta_2 = 180 - 2 \times 60 = 60^\circ \text{ C.W.}$$

$$\delta_3 = 60 - 30 = 30^\circ \text{ A.C.W.}$$

$$\delta_{\text{net}} = 30 + 60 - 30$$

$$\delta_{\text{net}} = 60^\circ$$

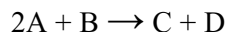
SUBJECT : CHEMISTRY

SECTION-A

53. **Ans (1)**

NCERT Based

69. **Ans (4)**



Rate of reaction =

$$\frac{-1}{2} \frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{d[C]}{dt} = \frac{d[D]}{dt}$$

Let rate of reaction = $k[A]^a[B]^b$

$$\text{Or, } \frac{d[C]}{dt} = k[A]^a[B]^b$$

Using the data given

$$1.2 \times 10^{-3} = k[0.1]^a[0.1]^b \quad \dots(i)$$

$$1.2 \times 10^{-3} = k[0.1]^a[0.2]^b \quad \dots(ii)$$

$$2.4 \times 10^{-3} = k[0.2]^a[0.1]^b \quad \dots(iii)$$

From equation (i) and (ii)

$$\frac{1.2 \times 10^{-3}}{1.2 \times 10^{-3}} = \frac{k[0.1]^a[0.1]^b}{k[0.1]^a[0.2]^b} \Rightarrow b = 0$$

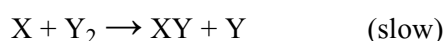
From equation (i) and (iii)

$$\frac{1.2 \times 10^{-3}}{2.4 \times 10^{-3}} = \frac{k[0.1]^a[0.1]^b}{k[0.2]^a[0.1]^b} \Rightarrow a = 1$$

$$\text{Thus, } \frac{d[C]}{dt} = k[A]^1[B]^0$$

70. **Ans (3)**

Note : Correct the reactions given in question as



Slow step is the rate determining step.

$$\text{Rate} = k[X][Y_2] \quad \dots(i)$$

$$\text{Equilibrium constant for fast step, } K = \frac{[X]^2}{[X_2]}$$

$$[X] = \sqrt{K[X_2]}$$

By substituting $[X]$ in equation (i), we get

$$\text{Rate} = k\sqrt{K[X_2]}[Y_2] = k'[X_2]^{1/2}[Y_2]$$

$$\therefore \text{Order of reaction} = \frac{1}{2} + 1 = \frac{3}{2} = 1.5$$

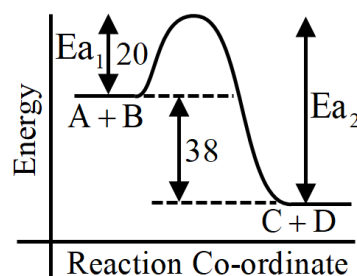
71. **Ans (4)**

For zero order reaction $C_0 - C_t = Kt$

If $C_t = 0.5 \text{ M}$; $t = 25 \text{ sec}$; $K = 2 \times 10^{-2}$ then
 $C_0 = 1.0 \text{ M}$

73. **Ans (4)**

$$E_{a2} = 58$$



74. **Ans (1)**

For zero order $t_{1/2} \propto [\text{conc.}]^0$

75. **Ans (3)**

For nth order reaction $r = K[A]^n$

$$K = \frac{r}{[A]^n} = \frac{(\text{Molar/s})}{(\text{Molar})^n} = (\text{Molar})^{1-n} \text{ s}^{-1} = \left(\frac{\text{mol}}{\text{L}}\right)^{1-n} \text{ s}^{-1} = \text{mol}^{1-n} \text{ L}^{n-1} \text{ s}^{-1}$$

76. **Ans (3)**

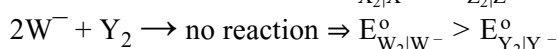
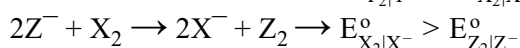
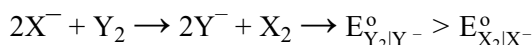
For an elementary reaction, order of reaction = stoichiometric coefficient.

78. **Ans (2)**

$$[A]_0 - [A] = kt$$

$$K = \frac{[A]_0 - [A]}{t}$$

79. **Ans (4)**



SECTION-B

93. **Ans (2)**

$$K = A \cdot e^{-E_a/RT}$$

$$\text{Given } K = A \cdot e^{-1000}$$

$$\frac{E_a}{RT} = 1000$$

94. **Ans (2)**

The value of rate constant of a pseudo first order reaction depends on temperature as well as on concentration of reactant present in excess, e.g., hydrolysis of ethyl acetate.

95. **Ans (2)**

Read in integrated rate equation for gas phase reactions of first order in our book.

96. **Ans (4)**

$k = Ae^{-E_a/RT}$. As E_a increases, $-E_a/RT$ decreases and so does k . When E_a decreases, k will increase exponentially. As T increases, E_a/RT decreases or $-E_a/RT$ increases and so does k .

97. **Ans (2)**

$$\alpha = \frac{\lambda_m}{\lambda_m^\infty}$$

$$K_a = \frac{C\alpha^2}{(1-\alpha)}$$

$$= \frac{C \times \lambda_m^2}{\lambda_\infty^2 \left(1 - \frac{\lambda_m}{\lambda_\infty}\right)}$$

$$= \frac{C\lambda_m^2}{\lambda_m^\infty (\lambda_m^\infty - \lambda_m)}$$

100. **Ans (3)**

SRP

X 0.65 V SRP : $Z > X > Y$

Y 1.40 V

Z -0.36 V

SUBJECT : BIOLOGY-I

SECTION-A

101. **Ans (3)**

NCERT-XII, Pg. # 57,59,60,62

103. **Ans (1)**

NCERT XII, Page # 64

106. **Ans (3)**

NCERT

Proliferative phase - Development of tertiary follicle

Secretory phase - Formation of corpus luteum

Ovulatory phase - Release of primary oocyte

107. **Ans (2)**

NCERT Pg#54, Module Pg#56

Oxytocin cause strong contraction during parturition.

110. **Ans (3)**

NCERT-XII, Pg. # 61/68(H) Fig: 4.4(b)

112. **Ans (2)**

NCERT (XII) Pg. # 64

113. **Ans (2)**

NCERT Pg. # 58

116. **Ans (3)**

NCERT XII Pg. # 153

119. **Ans (3)**

NCERT-XII Pg # 151, Para-8.2.2

132. **Ans (2)**

NCERT(XII) Pg#127 Para:7.1

133. **Ans (3)**

NCERT XIIth Pg.#127

135. **Ans (2)**

NCERT(XII) Pg#127 Para:7.1

SECTION-B

136. **Ans (3)**

NCERT Pg. # 61

137. **Ans (1)**

NCERT Pg. # 62

139. **Ans (1)**

NCERT-XII Pg. # 45/48(H) Fig:3.3(b)

141. **Ans (1)**

NCERT (XII) Pg. # 64

149. **Ans (2)**

NCERT XII, Pg.# 131

SUBJECT : BIOLOGY-II

SECTION-A

151. **Ans (2)**

NCERT (XII) Pg. # 130, fig 7.2

153. **Ans (1)**

NCERT (XIIth) Pg. # 139

159. **Ans (3)**

NCERT (XII) Pg. # 134, 7.7

164. **Ans (2)**
NCERT- Pg. # 277
165. **Ans (4)**
NCERT (XII) Pg. # 280
166. **Ans (4)**
NCERT (XIIth) Pg. # 275
167. **Ans (1)**
NCERT (XIIth) Pg. # 274
169. **Ans (4)**
NCERT XII pg.# 283
172. **Ans (3)**
NCERT (XII) Pg. # 282, 276, 277
175. **Ans (4)**
NCERT [XII], Pg # 28
176. **Ans (2)**
NCERT [XII], Pg # 21
177. **Ans (2)**
NCERT [XII], Pg # 21
178. **Ans (2)**
NCERT [XII], Pg # 22
179. **Ans (1)**
NCERT [XII] Pg # 23/25
180. **Ans (2)**
NCERT [XII], Pg # 26
181. **Ans (3)**
NCERT [XII] Pg # 26, Fig. (2.8)

182. **Ans (2)**
NCERT [XII] Pg # 25
183. **Ans (4)**
NCERT [XII] Pg # 25, Fig. - 2.7 (d)
184. **Ans (4)**
NCERT [XII] Pg # 30
185. **Ans (2)**
NCERT [XII] Pg # 31

SECTION-B

187. **Ans (3)**
NCERT XII Page No. # 134, 135
188. **Ans (4)**
NCERT-XII Pg# 133, Figure-7.6
189. **Ans (2)**
NCERT XII (E) Pg # 138, (H) Pg # 149
190. **Ans (3)**
NCERT Pg. # 272
191. **Ans (2)**
NCERT (XII) Pg. # 274
196. **Ans (4)**
NCERT-[XII], Pg # 30
197. **Ans (3)**
NCERT [XII], Pg # 21
199. **Ans (1)**
NCERT [XII] Pg # 26